



Koneru Lakshmaiah Education Foundation

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DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED BACKEND DEVELOPMENT (25CS1302E)

Y25 - 2026-2027 Trimester - 4

Project Abstract Submission Form

Sec No-10	Team No-29	Project Title- Sign Language Translator App
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Abstract

SignBridge: Intelligent Two-Way Sign Language Translator

Communication barriers between sign language users and the wider community remain a significant accessibility challenge, especially when interpreters are unavailable. **SignBridge** is an AI-powered, two-way Sign Language Translation platform that enables real-time communication by converting hand gestures into **text and speech**, while also translating **text or voice into visual sign representations**.

The system uses **OpenCV and MediaPipe** to capture hand gestures and extract hand landmarks, which are classified using a **TensorFlow/Keras or TensorFlow Lite** model. A **confidence-scoring mechanism** identifies uncertain predictions and asks users to repeat unclear gestures, improving translation reliability. Recognized signs are combined through a **Smart Phrase Builder** to generate meaningful sentences. The platform also includes **Emergency Mode** for critical phrases and a **Teach the AI** feature that allows users to add personalized gestures and expand the system's vocabulary.

The application is built using **React.js, FastAPI, and MySQL**, with **SQLAlchemy** for database integration and **JWT-based authentication** and **role-based access control** for security. A modular, service-oriented architecture with **Docker-based deployment, API documentation, testing, and monitoring** provides scalability and maintainability. By combining **AI-based recognition, confidence-aware translation, personalized learning, two-way communication, and modern backend engineering**, SignBridge aims to provide a reliable, adaptive, and accessible communication solution rather than a conventional gesture-detection system.

1. Team Details

Roll Number(s)	Name of the Student	Signature
2520090209	K.Hasini	
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2. Frameworks / Platforms / Software Used

- **Frontend:** React.js, HTML, CSS, JavaScript
- **Backend:** FastAPI (Python), REST APIs
- **AI/ML:** TensorFlow/Keras
- **Computer Vision:** OpenCV, MediaPipe
- **Database:** MySQL, SQLAlchemy
- **Authentication:** JWT, OAuth2
- **Speech:** Web Speech API / Text-to-Speech
- **API Documentation:** Swagger / OpenAPI
- **Testing:** Pytest
- **Deployment:** Docker, Docker Compose
- **Version Control & CI/CD:** Git, GitHub, GitHub Actions

Guide Name and Signature (remarks if any) _____